

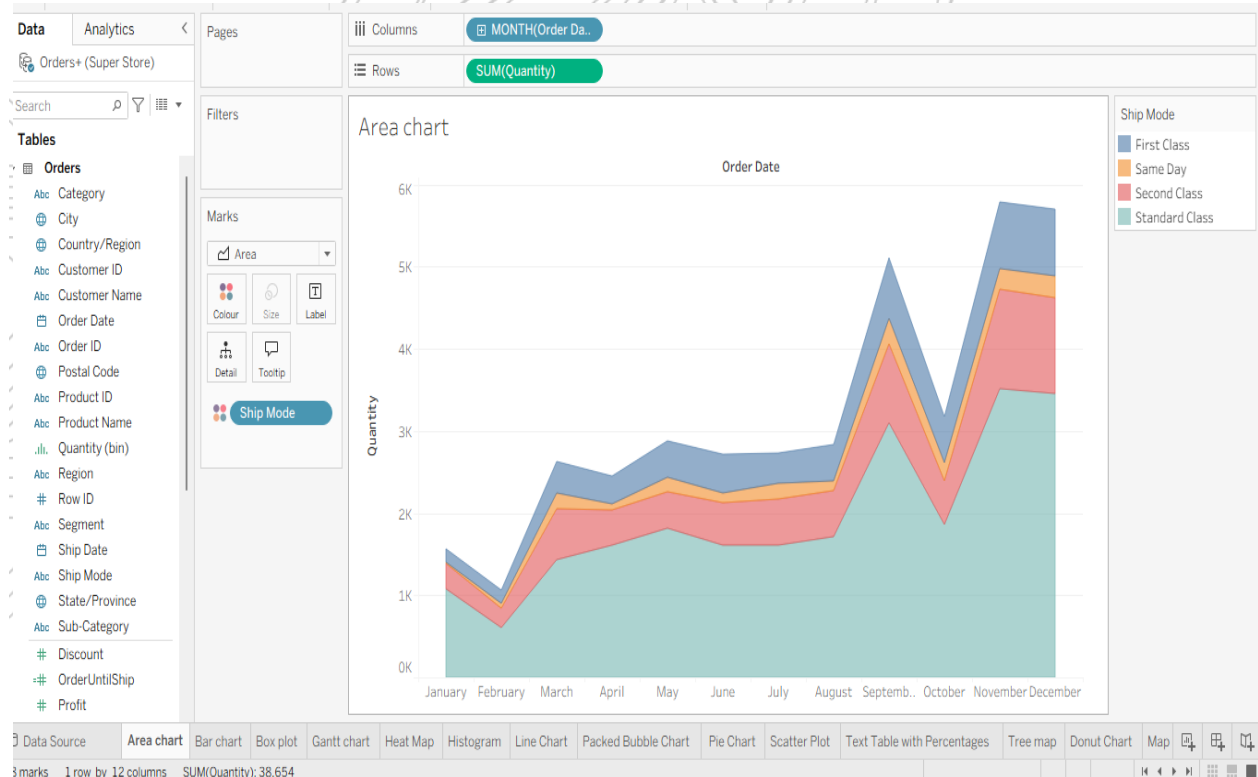
ASSIGNMENT

HALL TICKET NO	0225046065
NAME OF THE STUDENT	KOTAKONDA ANIL
LEARNING TRACK	DATA ANALATICS WITH TABLEAU

Sheet – 01: Area chart

1. From the **Data** pane, drag **Order Date** to the **Columns** shelf.
2. Click the drop-down arrow on **YEAR(Order Date)** in the Columns shelf and select **Month**.
3. Drag **Quantity** to the **Rows** shelf.
4. Drag **Ship Mode** to **Color** on the **Marks** card.
5. On the **Marks** card, click the chart type drop-down menu.
6. Select **Area** as the visualization type.

Out put:



Sheet – 02: Bar Chart

1. Place **Sales** on the **Rows** shelf to visualize total sales values.
 - Tableau automatically calculates the measure as **SUM(Sales)**.
 - If **Order Date** is present in the view, Tableau may initially display a line chart showing sales over time.
2. Go to the **Marks** card and open the chart type selector.
3. Choose **Bar** to convert the view into a bar chart.

Differentiate by Ship Mode

4. Drag **Ship Mode** to the **Color** option on the Marks card to assign a unique color to each shipping category.

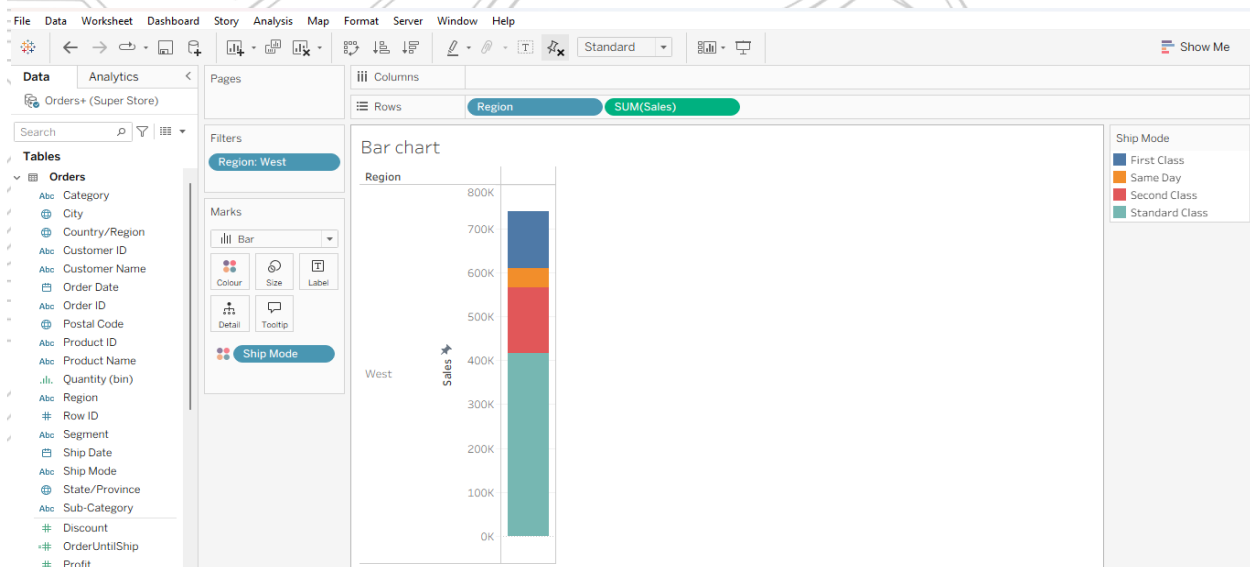
Display Sales by Region

5. Drag **Region** to the **Rows** shelf and position it before **Sales** to break down sales by region.

Show Only the West Region

6. Drag **Region** to the **Filters** shelf.
7. In the filter window, select **West** from the list of available regions.
8. Click **OK** to apply the filter and update the visualization.

Out Put:



Sheet – 03: Box Plot

Build the Initial View

1. Drag **Region** to the **Columns** shelf.
2. Drag **Profit** to the **Rows** shelf.

Modify the Measure Calculation

3. Click the drop-down arrow on **Profit** in the Rows shelf.
4. Select **Measure** from the menu.
5. Choose **Median** as the aggregation method.

Add a Category Breakdown

6. Drag **Category** to the **Columns** shelf.
7. Place it to the right of **Region** to compare profit distributions across product categories.

Create the Box Plot

8. Open the **Show Me** panel from the toolbar.
9. Select **Box-and-Whisker Plot** to transform the visualization.

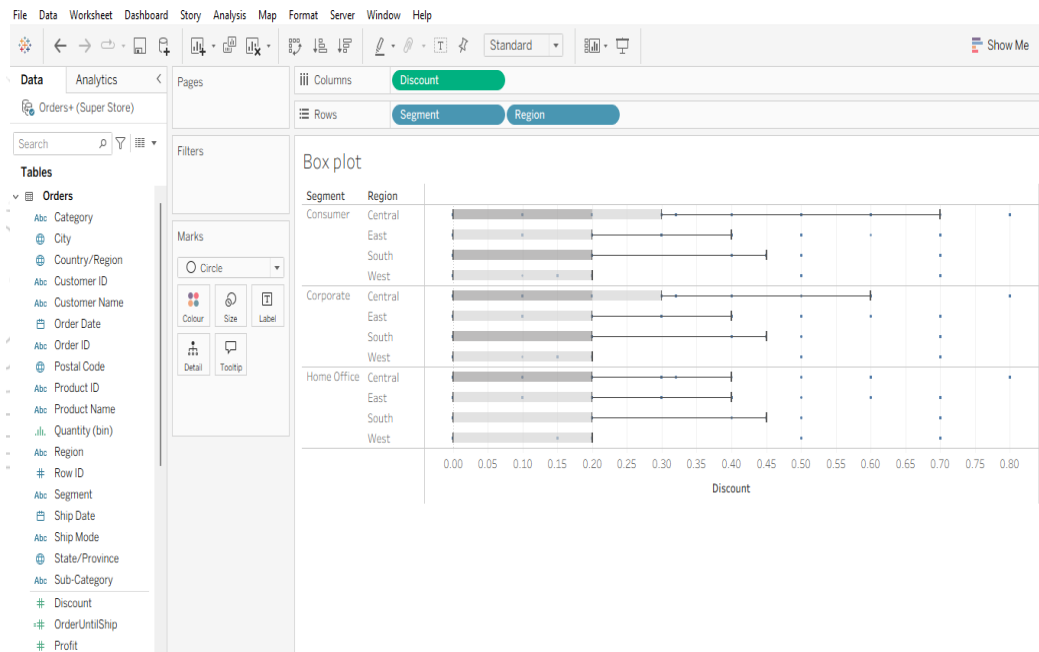
Refine the Layout

10. Ensure **Category** and **Region** remain on the **Columns** shelf in the correct order.
11. From the **Analysis** menu, review the data settings.
12. Disable **Aggregate Measures** to display the full distribution of records.

Change Chart Orientation

13. Click the **Swap Rows and Columns** button to display the box plots horizontally.

Out put:



Sheet – 04: Gantt Chart

Step 1: Create a Time-Based View

1. Drag **Order Date** from the Data pane to the **Columns** shelf.
2. Tableau automatically displays the date at the **Year** level.
3. Click the drop-down menu on the date field and select **Week Number** to view data by week.

Step 2: Add Dimensions to the Rows Shelf

4. Drag **Sub-Category** to the **Rows** shelf.
5. Drag **Ship Mode** to the **Rows** shelf and place it beside **Sub-Category**.
6. This organizes the data by product sub-category and shipping method.

Step 3: Convert the Visualization to a Gantt Chart

7. Open the **Marks** card.
8. From the mark type drop-down menu, choose **Gantt Bar**.

Step 4: Create a Shipping Delay Calculation

9. Select **Analysis** → **Create Calculated Field**.
10. Name the field **OrderUntilShip**.
11. Enter the following formula:

```
DATEDIFF('day', [Order Date], [Ship Date])
```

12. Click **OK** to save the calculation.

Step 5: Use the Calculation to Size the Bars

13. Drag **OrderUntilShip** to the **Size** shelf on the Marks card.
14. The length of each Gantt bar now represents the shipping delay in days.

Step 6: Display Average Shipping Delay

15. Right-click **SUM(OrderUntilShip)** on the Marks card.
16. Select **Measure** → **Average** to show the average delay instead of the total.

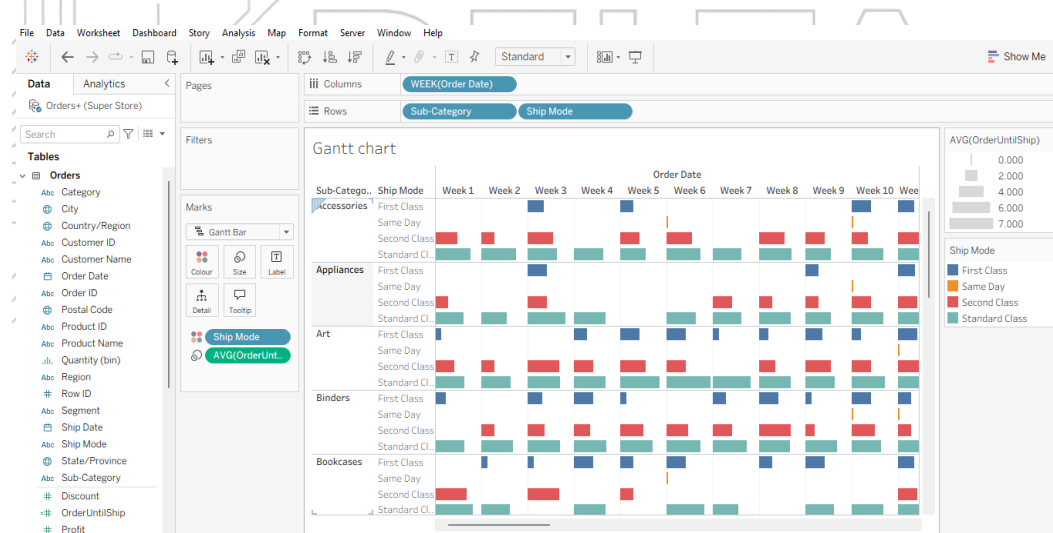
Step 7: Differentiate Ship Modes with Colors

17. Drag **Ship Mode** to **Color** on the Marks card.
18. Tableau assigns a unique color to each shipping method for easy comparison.

Step 8: Analyze the Results

19. Review the weekly timeline to compare shipping delays across sub-categories and ship modes.
20. Adjust the bar size and formatting as needed for better readability.

Out put:



Sheet 5: Heat Map

Step 1: Drag Segment to the Columns shelf.

Step 2: Drag Region to the Rows shelf.

Step 3: Drag Sub-Category to the Rows shelf (place it to the right of Region).

Step 4: Drag Profit to Colour on the Marks card.

Step 5: Click the Colour button on the Marks card, and under Border, choose Medium Gray.

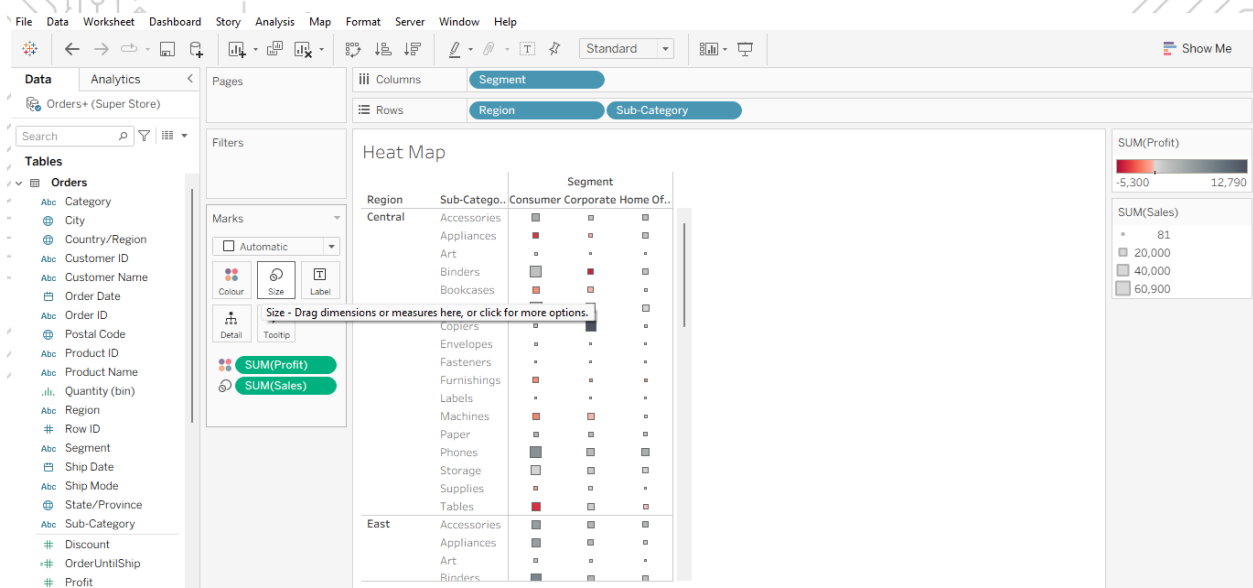
Step 6: On the Profit colour legend (top right), click the drop-down arrow and select Edit Colours.

Step 7: Change the Palette to Red-Black Diverging, check Use Full Colour Range, and click OK.

Step 8: On the Marks Card, click the drop-down menu and change the mark type from Automatic to Square.

Step 9: Drag Sales to Label on the Marks card to display the actual data values inside each colored square.

Out put:



Sheet 6 : Histogram

Step 1: Drag Quantity to the Columns shelf.

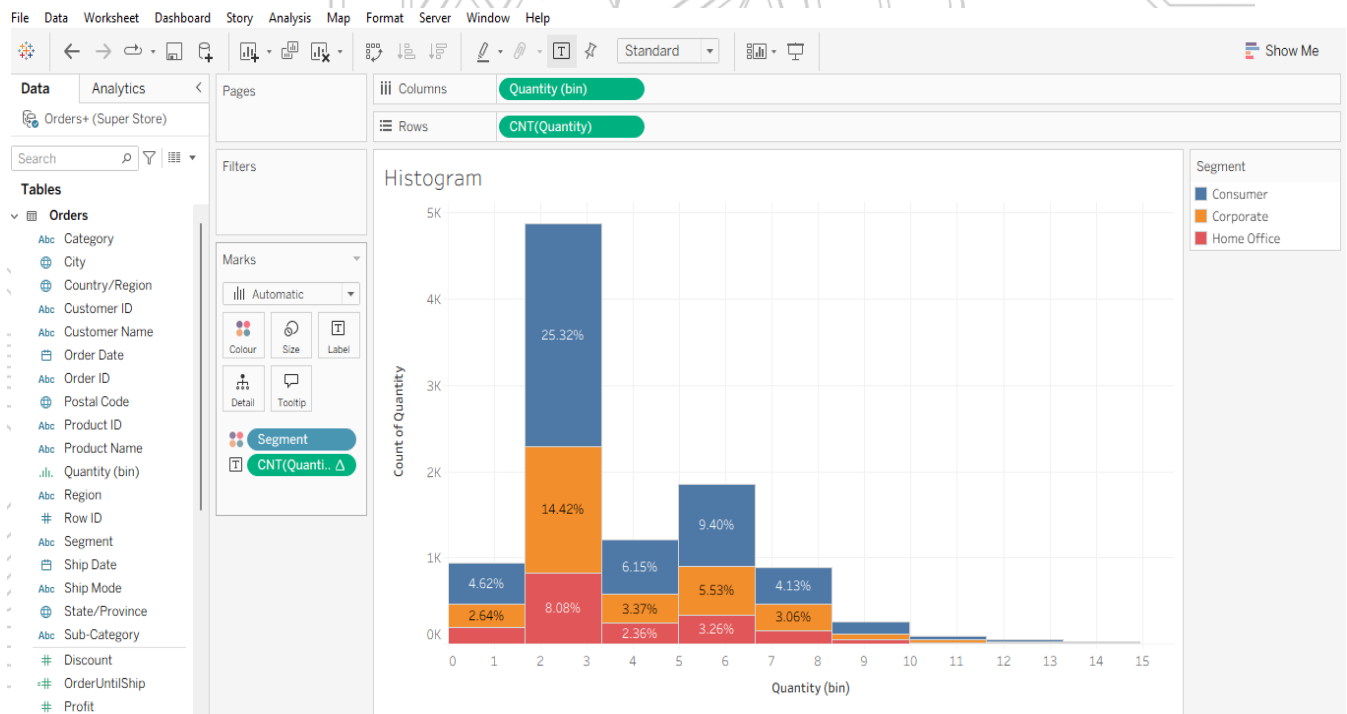
Step 2: Click Show Me (top right corner) and select the Histogram icon.

Step 3: Drag Segment to the Columns shelf (place it to the left of the Quantity (bin) pill).

Step 4: Hold Ctrl (or Cmd on Mac), click the CNT(Quantity) pill on the Rows shelf, and drag a copy of it onto Label on the Marks card.

Step 5: Right-click the CNT(Quantity) pill inside the Marks card and select Quick Table Calculation right arrow Percent of Total.

Out put:



Sheet 7: Line Chart

Step 1: Drag Ship Date to the Columns shelf.

Step 2: Drag Sales to the Rows shelf.

Step 3: Drag Profit to the Rows shelf (place it to the right of Sales).

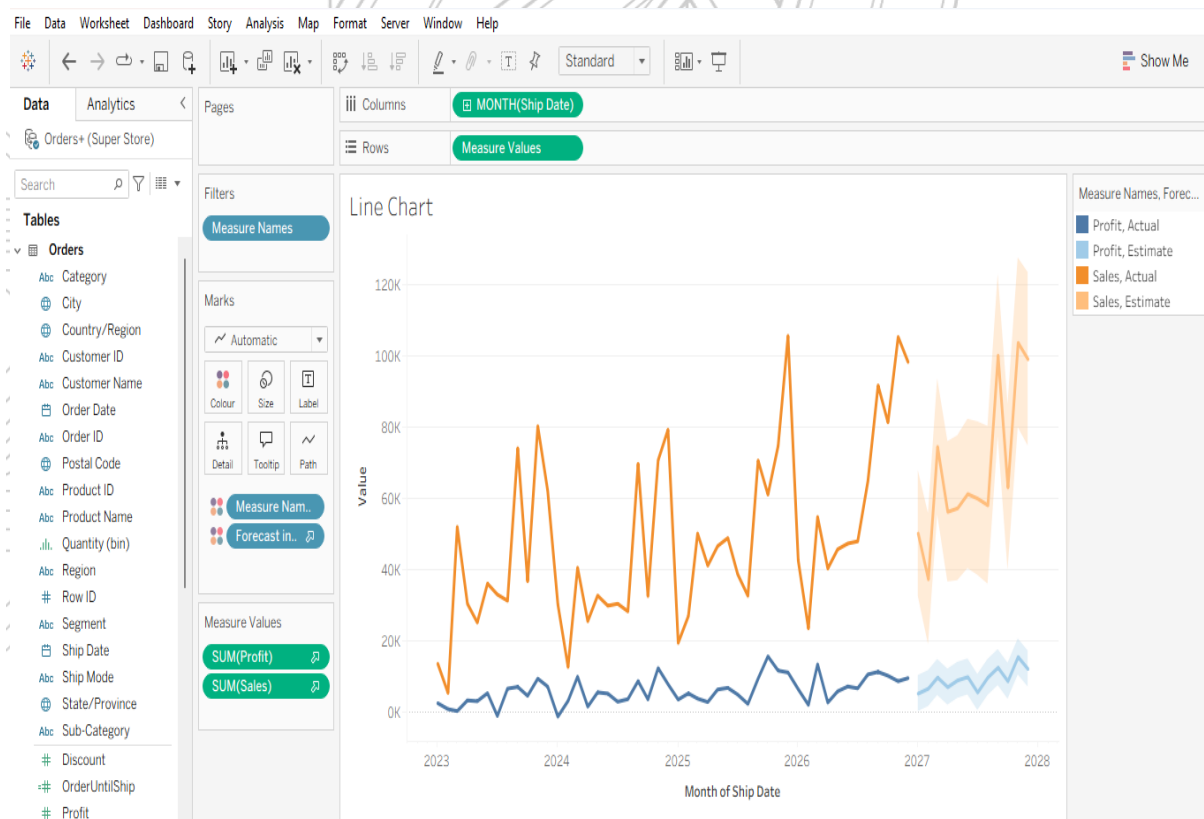
Step 4: Right-click the SUM(Profit) pill on the Rows shelf and select Dual Axis, then right-click the right axis on the chart and select Synchronize Axis.

Step 5: Click the drop-down arrow on the YEARS (Ship Date) pill on the Columns shelf and select Month (from the lower continuous section).

Step 6: Switch from the Data Pane to the Analytics Pane on the left side of the screen.

Step 7: Drag Forecast and drop it directly into the chart area.

Out put:



Sheet 8: Packed Bubble chart

Step 1: Drag Category to the Columns shelf.

Step 2: Drag Sales to the Rows shelf.

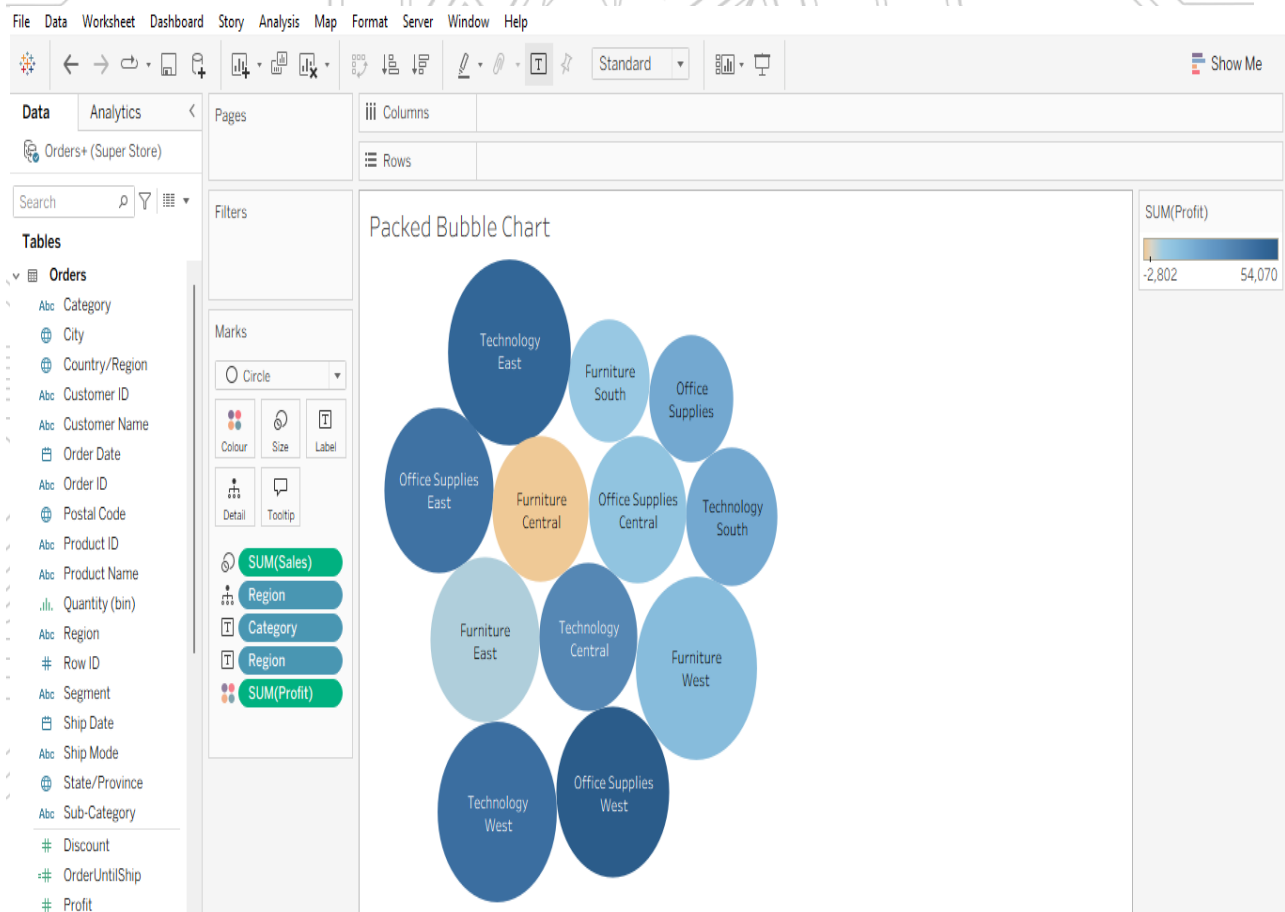
Step 3: Click Show Me (top right corner) and select the Packed Bubbles icon.

Step 4: Drag Region to Detail on the Marks card.

Step 5: Drag Category to Color on the Marks card.

Step 6: Drag Region to Label on the Marks card.

Out put:



Sheet 9 : Pie chart

Step 1: Drag Profit to the Columns shelf.

Step 2: Drag Category to the Rows shelf.

Step 3: Click Show Me (top right corner) and select the Pie Chart icon.

Step 4: On the top toolbar, click the Fit drop-down menu (which usually says Standard) and select Entire View.

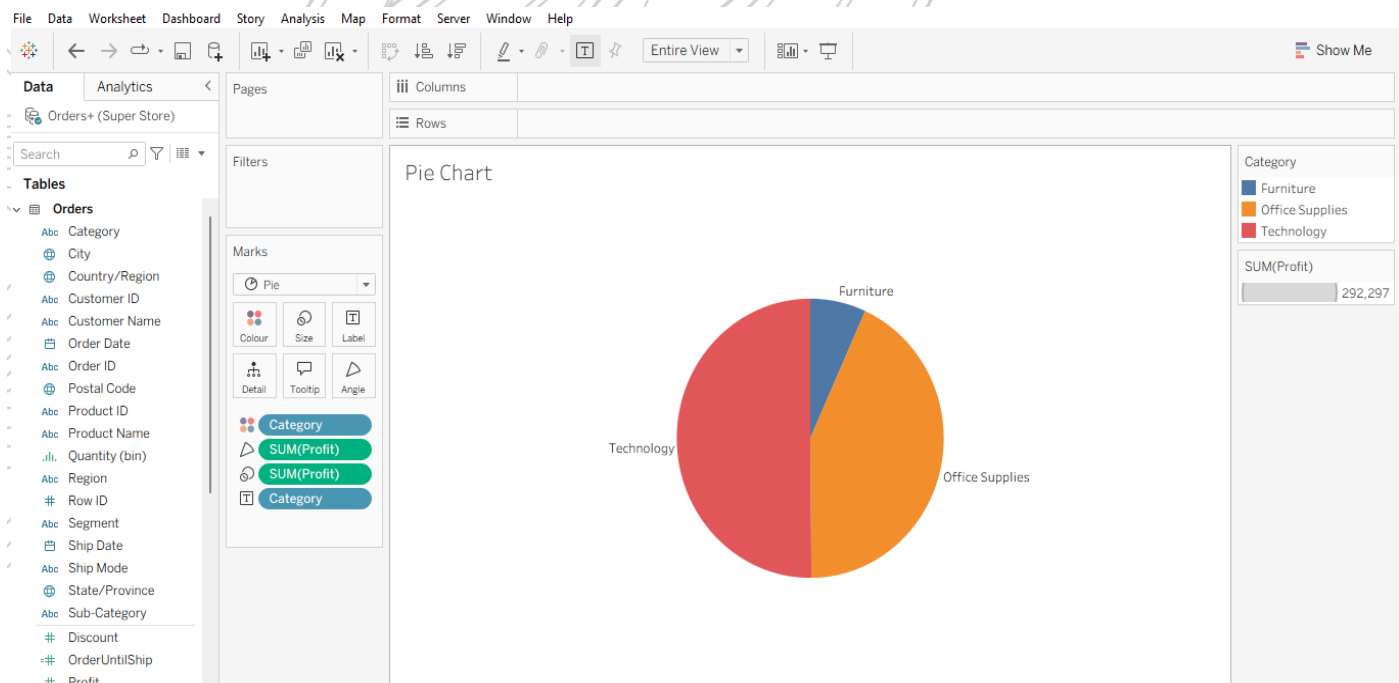
Step 5: Double-click an empty space in the Rows shelf, type "MIN(0)", and press Enter. Repeat this step to add a second "MIN(0)" right next to it in the Rows shelf.

Step 6: Right-click the second "MIN(0)" pill on the Rows shelf and select Dual Axis.

Step 7: On the Marks card, click on the second "MIN(0)" section, remove Category and Profit from it, change the colour to White, and slightly decrease its Size slider.

Step 8: Drag Profit to Label on the second "MIN(0)" Marks card to display the total profit right in the Center of the donut chart.

Out put:



Sheet 10 : Scatter Plot

Step 1: Drag Sales to the Columns shelf.

Step 2: Drag Profit to the Rows shelf.

Step 3: Drag Category to Colour on the Marks card.

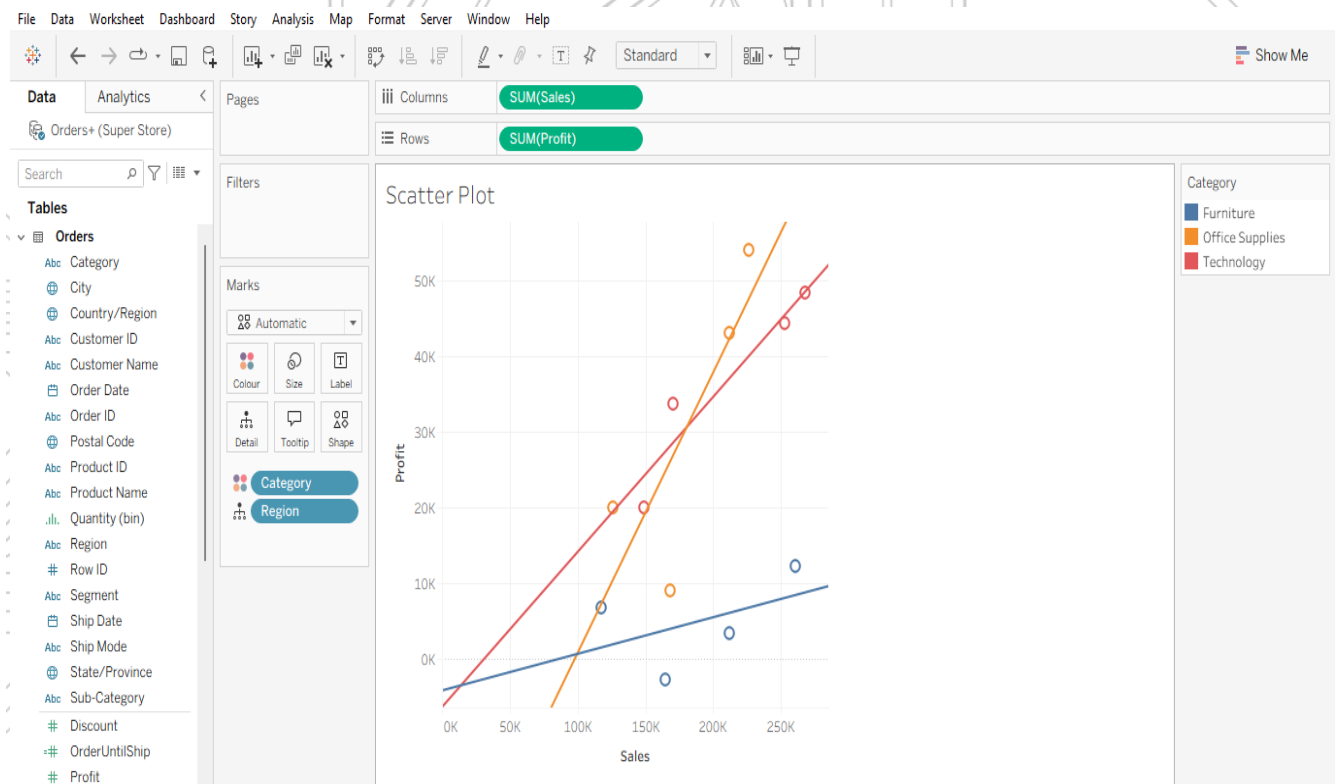
Step 4: Drag Region to Detail on the Marks card.

Step 5: Switch from the Data Pane to the Analytics Pane on the left side of the screen.

Step 6: Drag Average Line and drop it onto the Table option under the Sales (X-axis) section.

Step 7: Drag Average Line again and drop it onto the Table option under the Profit (Y-axis) section.

Out put:



Sheet 11 : Text Table With Percentages

Step 1: Drag Order Date to the Columns shelf.

Step 2: Drag Sub-Category to the Rows shelf.

Step 3: Drag Sales to Text on the Marks card.

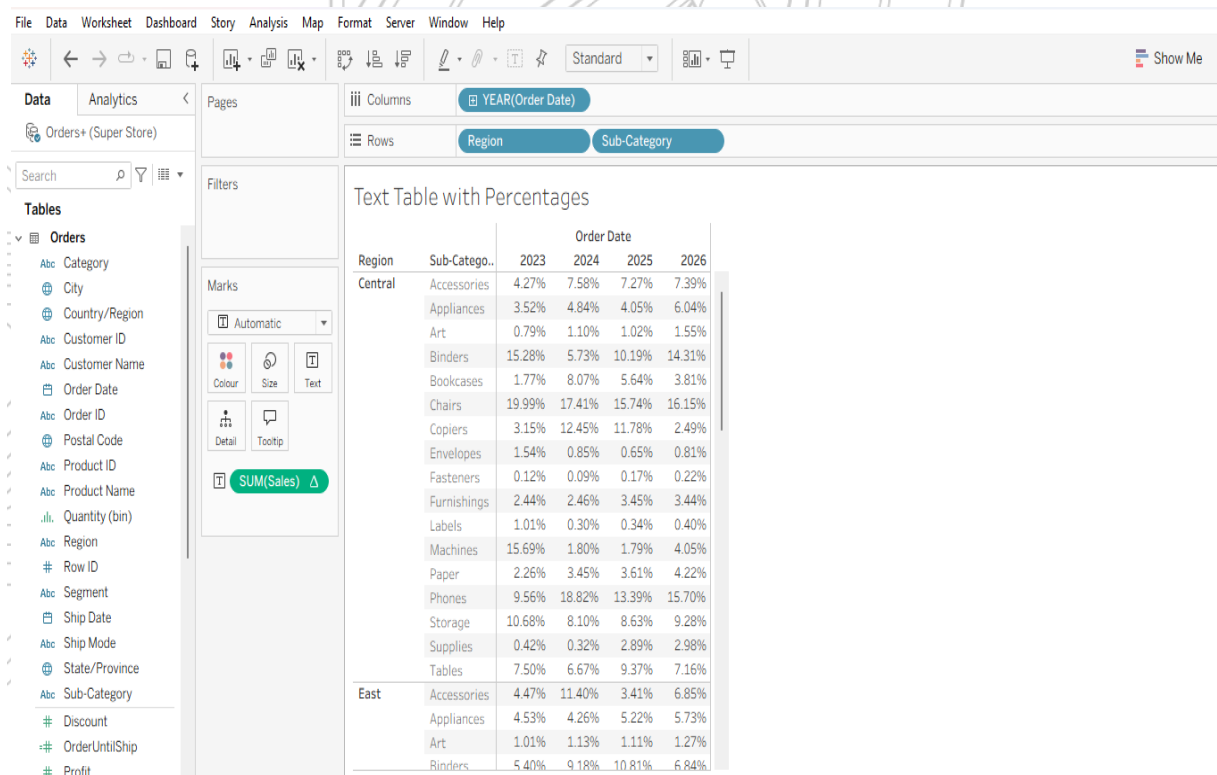
Step 4: Drag Region to the Rows shelf (place it to the left of Sub-Category).

Step 5: Drag Sales from the Data pane a second time and drop it directly onto Text on the Marks card.

Step 6: Right-click the second SUM(Sales) pill on the Marks card and select Add Table Calculation.

Step 7: In the configuration window, set the Calculation Type to Percent of Total, set Compute Using to Pane (Down), and close the box.

Out put:



The screenshot shows the Tableau Desktop interface with the following configuration:

- Columns:** YEAR(Order Date)
- Rows:** Region, Sub-Category
- Marks:** SUM(Sales), SUM(Sales)
- Table Calculation:** The second SUM(Sales) pill is configured with Calculation Type: Percent of Total, Compute Using: Pane (Down).

The resulting text table is titled "Text Table with Percentages" and displays the following data:

		Order Date			
Region	Sub-Catego..	2023	2024	2025	2026
Central	Accessories	4.27%	7.58%	7.27%	7.39%
	Appliances	3.52%	4.84%	4.05%	6.04%
	Art	0.79%	1.10%	1.02%	1.55%
	Binders	15.28%	5.73%	10.19%	14.31%
	Bookcases	1.77%	8.07%	5.64%	3.81%
	Chairs	19.99%	17.41%	15.74%	16.15%
	Copiers	3.15%	12.45%	11.78%	2.49%
	Envelopes	1.54%	0.85%	0.65%	0.81%
	Fasteners	0.12%	0.09%	0.17%	0.22%
	Furnishings	2.44%	2.46%	3.45%	3.44%
	Labels	1.01%	0.30%	0.34%	0.40%
	Machines	15.69%	1.80%	1.79%	4.05%
	Paper	2.26%	3.45%	3.61%	4.22%
	Phones	9.56%	18.82%	13.39%	15.70%
	Storage	10.68%	8.10%	8.63%	9.28%
East	Supplies	0.42%	0.32%	2.89%	2.98%
	Tables	7.50%	6.67%	9.37%	7.16%
	Accessories	4.47%	11.40%	3.41%	6.85%
	Appliances	4.53%	4.26%	5.22%	5.73%
	Art	1.01%	1.13%	1.11%	1.27%
	Binders	5.40%	9.18%	10.81%	6.84%

Sheet 12 : Tree map

Step 1: Drag Sub-Category to the Columns shelf.

Step 2: Drag Sales to the Rows shelf.

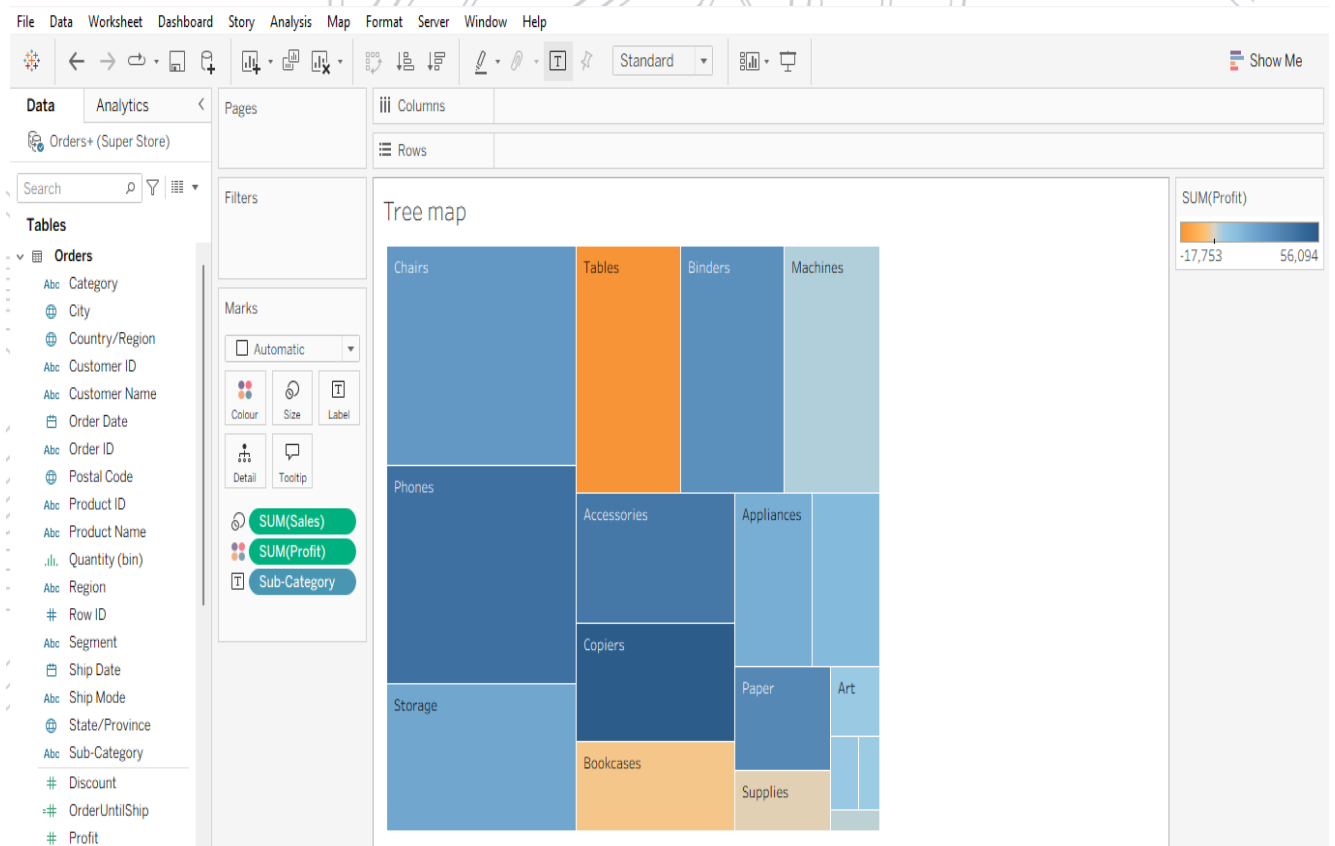
Step 3: Click Show Me (top right corner) and select the Tree map icon.

Step 4: Drag Category to Color on the Marks card.

Step 5: Drag Category to Label on the Marks card.

Step 6: Drag Sub-Category to Label on the Marks card to display both levels inside the blocks.

Out put:



Sheet 13 : Donut chart

Step 1: Click the top Analysis menu, select Create Calculated Field, name it “Zero“, type “0“ in the formula area, and click OK.

Step 2: Drag Zero to the Rows shelf.

Step 3: On the Marks Card, click the drop-down menu, select Pie, and then change the view setting on the top toolbar from Standard to Entire View.

Step 4: Click Size on the Marks card and move the slider to the right to make the pie larger.

Step 5: Drag Zero to the Rows shelf a second time (place it to the right of the first one).

Step 6: Right-click the second SUM(Zero) pill on the Rows shelf and select Dual Axis.

Step 7: On the Marks card, click the second SUM(Zero) section, click Size to make it slightly smaller, then click Color and select White.

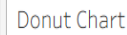
Step 8: Click the first SUM(Zero) section on the Marks card, then drag Region to Color, Region to Label, Sales to Label, and Sales to Angle.

Step 9: Right-click Region in the Data Pane, select Create right arrow Parameter, name it “Select Region“, and click OK. Right-click the new parameter and select Show Parameter Control.

Step 10: Create a new Calculated Field named “Selected Region Sales“, type the formula “IF [Region] = [Select Region] THEN [Sales] END“, click OK, and drag it to Label on the second SUM(Zero) Marks card.

The screenshot displays the Tableau Desktop interface. On the left, the 'Data' pane shows a hierarchy of tables: Orders, Customers, Regions, and Sales. The 'Columns' shelf contains 'SUM(Zero)' and the 'Rows' shelf contains 'SUM(Zero)'. The 'Marks' shelf is set to 'Pie'. The 'Filters' shelf is empty. The 'Legend' on the right shows the 'Region' dimension with four categories: Central (blue), East (orange), South (red), and West (teal). The main view shows a donut chart titled 'Donut Chart' representing 'Total Sales' by 'Region'. The chart is divided into four segments: Central (503,171), East (691,828), South (391,722), and West (739,814). The chart is set to 'Pie' and 'Total Sales' is the measure.

Region	Sales
Central	503,171
East	691,828
South	391,722
West	739,814



Region	Sales
Central	503,171
East	691,828
South	391,722
West	9,814

Region

- Central
- East
- South
- West

Sheet 14 : Map Chart

Step 1: Drag State from the Data pane and drop it directly onto the empty center workspace area.

Step 2: Drag Sales to Size on the Marks card.

Step 3: On the Marks Card, click the drop-down menu and change it from Automatic to Circle.

Step 4: Drag Profit to Color on the Marks card, click Edit Colors, choose the Orange-Blue Diverging palette, and click OK.

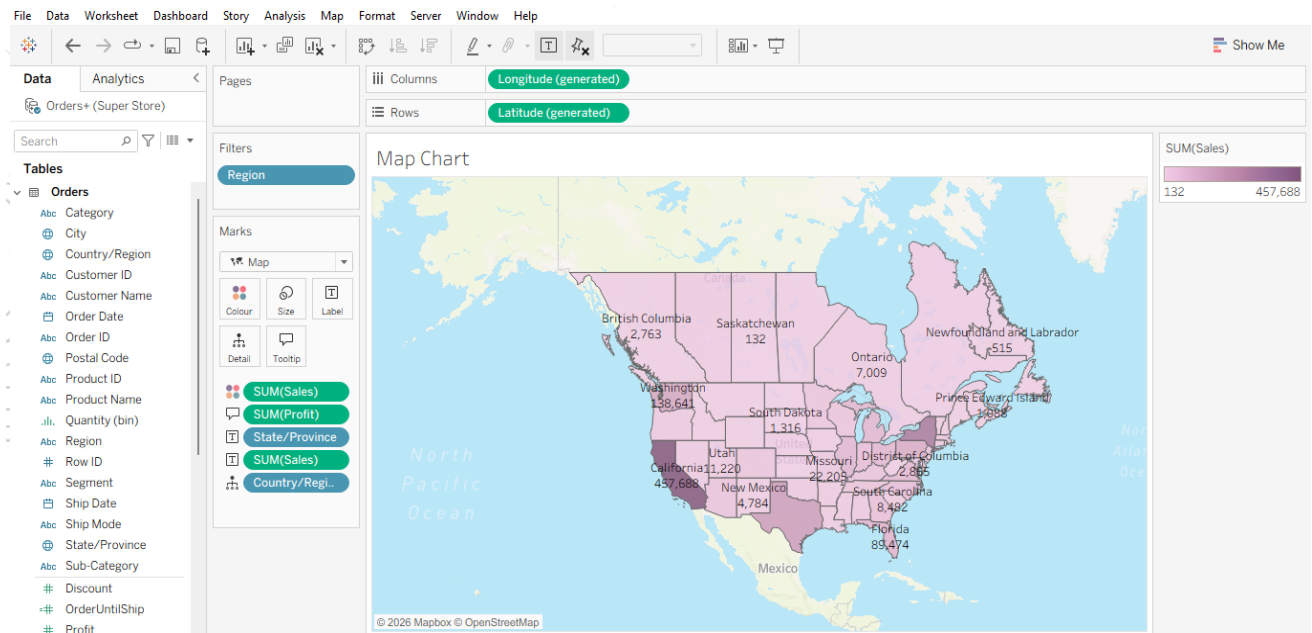
Step 5: Drag State to Label and Sales to Label on the Marks card.

Step 6: Drag Region to the Filters shelf, check All, and click OK.

Step 7: Click Size on the Marks card and adjust the slider to ensure the bubble overlaps are clearly readable.

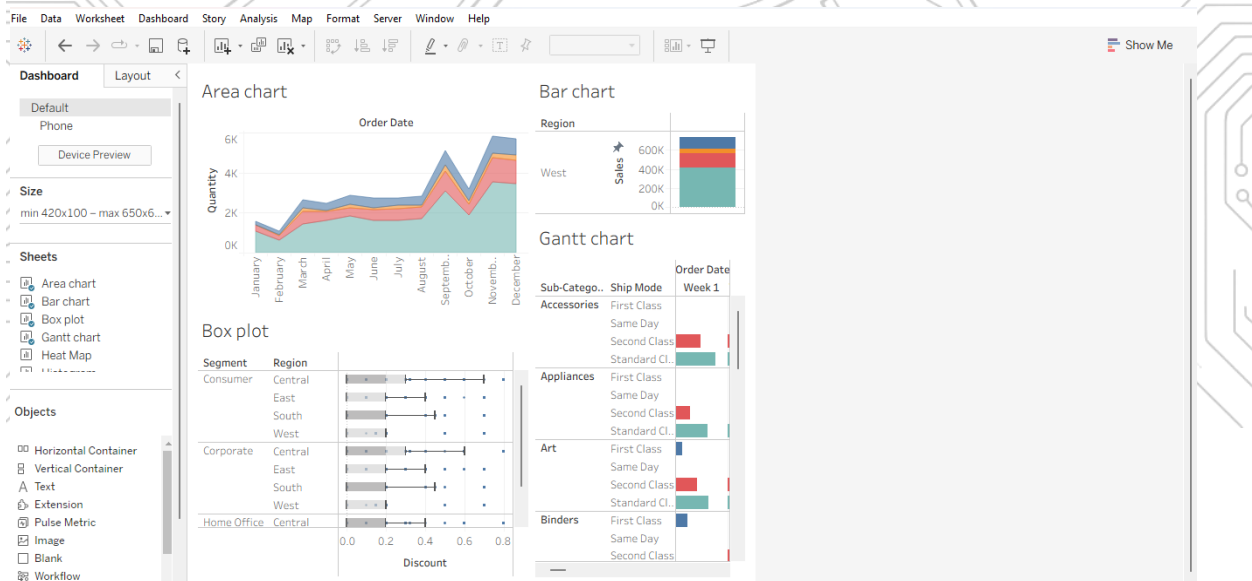
Step 8: Go to the top menu bar, click Map right arrow Map Layers, change the Background style to Streets, and set the Washout to 70%.

Out put:

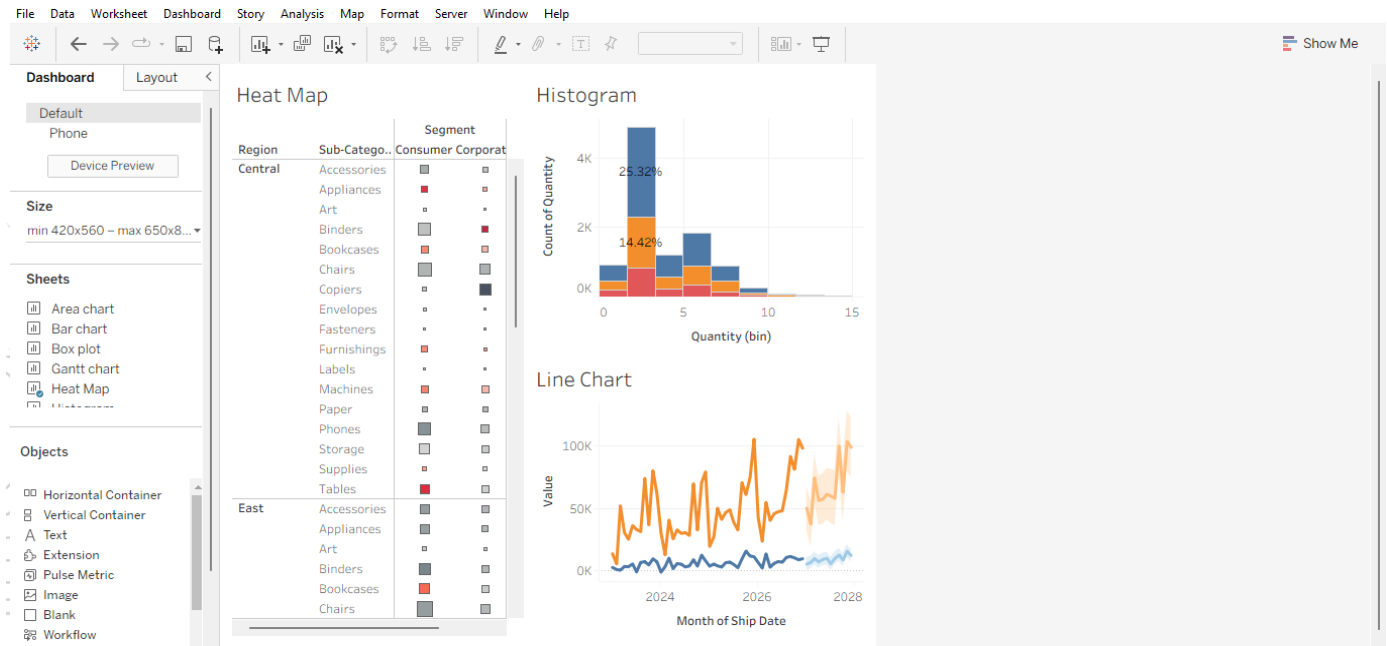


DASHBOARD

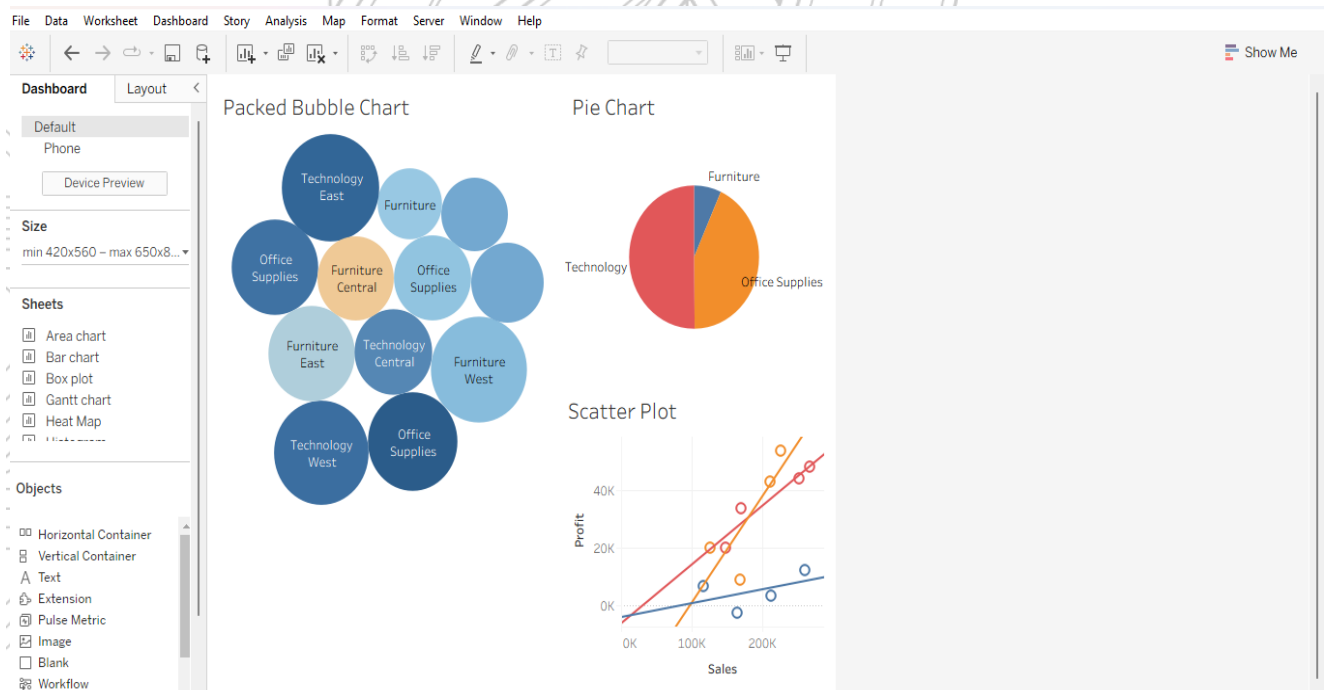
DASHBOARD-01



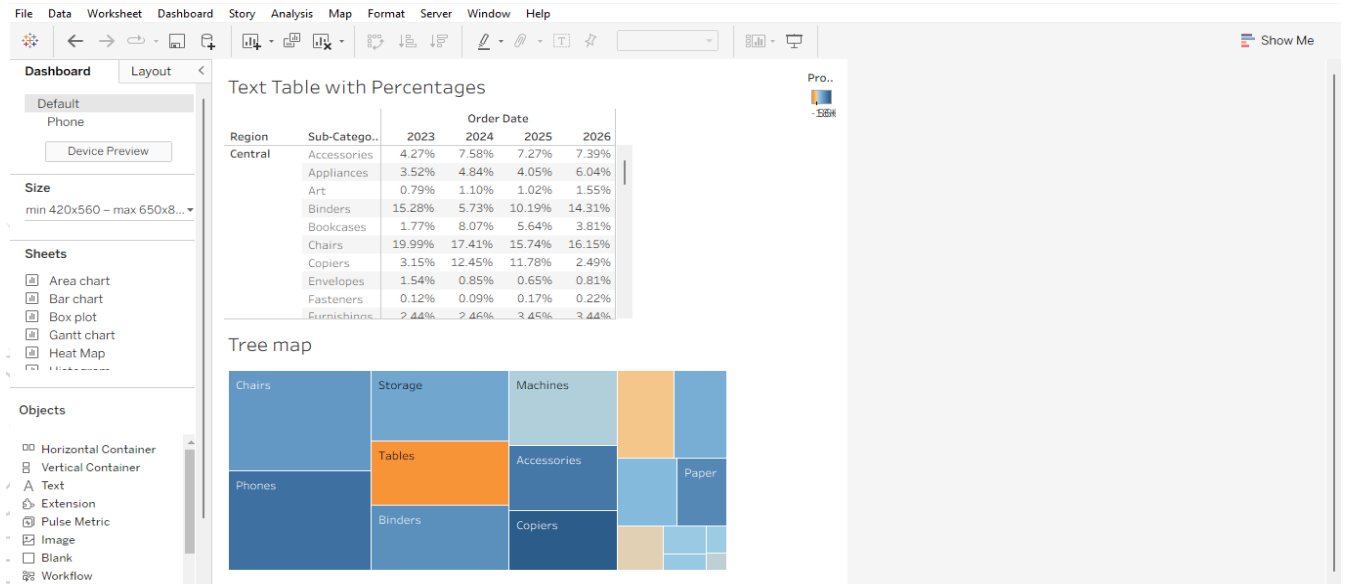
DASHBOARD-02



DASHBOARD-03



DASHBOARD-04



DASHBOARD-05

